

Real-Time Multi-Target Tracking Using YDLIDAR G2 and UKF-CTRV

EE4084 Project Presentation

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Project scope: LiDAR data acquisition, DBSCAN clustering, UKF-CTRV tracking, JPDA-based association, and NIS consistency analysis.

Presentation Roadmap

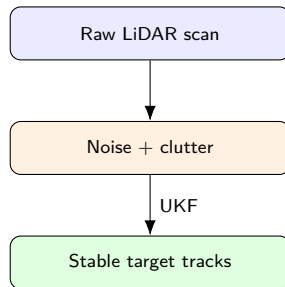
- 1 Motivation and project objective
- 2 System architecture and data flow
- 3 Process model: CTRV motion
- 4 Measurement model: LiDAR cluster centers
- 5 Filter design: UKF and association
- 6 Results, comparison, and demonstration



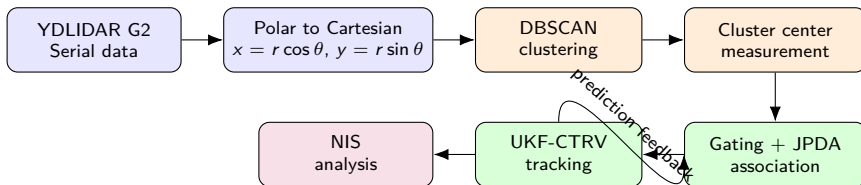
YDLIDAR G2 sensor used for real-time target tracking

Project Motivation

- Indoor robotic systems need reliable awareness of moving objects.
- Raw LiDAR points are noisy, dense, and not directly suitable for stable tracking.
- Human motion is nonlinear: targets may turn, stop, cross, or temporarily disappear.
- A nonlinear filter is needed to estimate smooth position, velocity, heading, and turn rate.



System Architecture



Main implementation idea

Each frame converts raw scan data into candidate target centers, then the UKF predicts and updates each active track in real time.

Preprocessing

- Minimum and maximum valid range filtering
- Background learning during the first frames
- Detection range limited to the indoor target area
- Invalid and isolated measurements removed

Clustering

DBSCAN groups dense point regions into candidate targets. Cluster width and height thresholds reject walls, noise, and unrealistic objects.

Parameter	Value
Frame duration	0.15 s
Detection range	3.5 m
DBSCAN ϵ	0.22 m
Minimum points	8
Maximum tracks	6
Gate radius	0.75 m

Process Model: CTRV Motion

State vector

$$\mathbf{x} = [p_x \quad p_y \quad v \quad \psi \quad \omega]^T$$

- p_x, p_y : target position
- v : linear velocity
- ψ : heading angle
- ω : turn rate

Nonzero turn rate

$$p_x^+ = p_x + \frac{v}{\omega} [\sin(\psi + \omega\Delta t) - \sin \psi]$$

$$p_y^+ = p_y + \frac{v}{\omega} [-\cos(\psi + \omega\Delta t) + \cos \psi]$$

$$\psi^+ = \psi + \omega\Delta t$$

For very small ω , the implementation switches to straight-line motion.

Measurement Model

LiDAR measurement

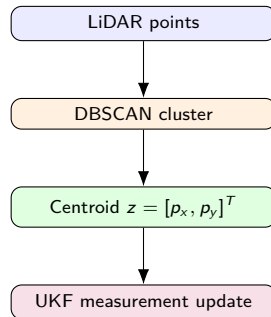
$$\mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{v}_k, \quad h(\mathbf{x}_k) = \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

The measurement is the 2-D center of the detected LiDAR cluster.

Noise model

$$\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, R), \quad R = \sigma_{meas}^2 I_2$$

In the implementation, $\sigma_{meas} = 0.035$ m.



Filter Design: Unscented Kalman Filter

- 1 Initialize state and covariance for each track.
- 2 Generate sigma points around the current estimate.
- 3 Propagate sigma points through the nonlinear CTRV model.
- 4 Compute predicted mean and covariance.
- 5 Predict measurement and innovation covariance.
- 6 Update state using the Kalman gain.

Why UKF?

- Handles nonlinear motion without Jacobian matrices.
- More stable than EKF under turning motion.
- Lower computational cost than particle filtering for this real-time setup.

Used noise terms

$$\sigma_{acc} = 3.5, \sigma_{\omega} = 0.8, \alpha = 10^{-3}, \beta = 2, \kappa = 0.$$

Data Association and Track Management

Association logic

- A predicted measurement is generated for every active track.
- Measurements outside the gate are rejected.
- JPDA-style association probabilities combine valid innovations.
- The best measurement updates the track history and visualization.

Track life cycle

- New track: unmatched measurement
- Confirmation: 2 successful hits
- Deletion: more than 6 missed frames
- Maximum active tracks: 6

Benefit

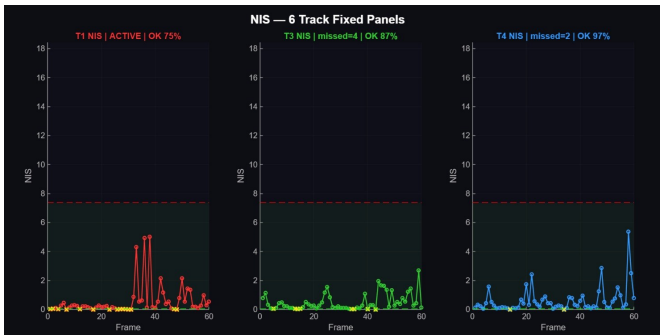
This prevents unstable tracks from appearing and keeps targets alive during short occlusions.

Consistency Analysis: NIS

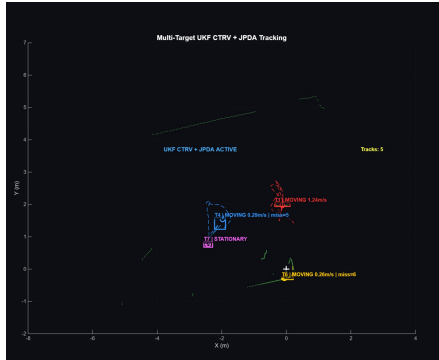
Normalized Innovation Squared (NIS)

$$\text{NIS} = \nu^T S^{-1} \nu$$

NIS checks whether the measurement innovation is statistically consistent with the predicted uncertainty. In this project, it is used to evaluate whether the UKF measurement update behaves reliably during real-time multi-target tracking.



Comparative Results and Observations



Real-time multi-target tracking result.

Approach	Observation
Raw DBSCAN centers	Detects targets but positions are jittery and can disappear under occlusion.
Linear KF	Too limited for nonlinear turning motion.
EKF	Requires Jacobians and may suffer from linearization errors.
Particle Filter	Strong nonlinear handling but heavier for real-time use.
UKF-CTRV	Smooth trajectory and velocity estimation with real-time execution.

System Demonstration Plan

What to show in the video

- 1 Connect YDLIDAR G2 and start MATLAB script.
- 2 Show background learning stage.
- 3 Move one or more targets in the sensor area.
- 4 Show bounding boxes, velocity arrows, target IDs, and track trails.
- 5 Show NIS panels and explain consistency briefly.

Narration focus

The demo should prove that the system does not only detect points; it maintains identities, estimates motion states, and handles short measurement losses.

Screen recording tip

Record the main tracking window and the NISpanel side by side.

Conclusion and Submission Comment

Conclusion

This project implements a real-time LiDAR-based multi-target tracking system using DBSCAN detection and UKF-CTRV state estimation. The results show smooth trajectories, stable velocity estimates, reduced false tracks, and statistically meaningful consistency monitoring.

Comment template for submission

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Project Title: Real-Time Multi-Target Tracking Using YDLIDAR G2 and Unscented Kalman Filter with CTRV Motion Model

YouTube Presentation Link: https://www.youtube.com/watch?v=TkG5Gu6K_jo